

EXECUTIVE REVIEW

Three Modules That Carry the Pipeline

Specialty-aware clinical risk cohort stack · endocrinology & cardiology

This stack converts linked clinical and claims data into prioritised, auditable outreach lists while holding protected health information inside enforced boundaries. Its design rests on one idea — risk is inferred from what is *absent* in the record, not only from abnormal values — and on a layered separation that keeps clinical computation, display, statistical gating, and model access independent and individually testable. The three modules below carry that value and the associated compliance risk.

1

Specialty Strategy Engine

`getClinicalFlags()` · `SPECIALTY_STRATEGIES`

Detects actionable care gaps from **claims absence**: an uncontrolled HbA1c with no medication-management or self-management-training claim, or atrial fibrillation with no anticoagulant on record. Each measure resolves to one of four explicit states — present, absent, or present-but-unparseable — so no value silently escapes evaluation. Each specialty is a self-contained registered strategy, so a new specialty is added by registration alone, with no change to the engine.

Value Surfaces patients invisible to value-only tools; gaps live in the claims stream, not the chart.

Integrity Missing vs. malformed data route to distinct codes (**MISSING** → outreach, **INVALID** → upstream fix); medication and education gaps flag independently; AFib matched case-insensitively across its full code family.

2

Stratification — Compute Before Display

`stratifyPatients()`

Assigns each patient a graded risk tier on **exact clinical values**, before any privacy transformation, and attaches their flags in the same pass. Tiers run *low* → *high* → *critical*; when the deciding measure is absent or unparseable the tier is **unknown**, never defaulted to low — an unmeasured patient is indeterminate, not safe.

Value One consistent, auditable risk view; tiering precision is preserved because it reads real measurements.

Integrity Malformed records guarded; thresholds read from one immutable config; an unsupported specialty fails loudly rather than silently producing an empty cohort.

3

Cohort Gating & Model Integration

`assessCohortEvidence()` · `assertAggregateOnly()` · `getLLMFeedback()`

Separates **evidence from reason**: a pure, testable assessment decides whether a cohort is precise and stable enough *before* any model call. It gates on the cohort estimate's 95% margin of error and on variance drift against a *supplied* historical baseline, and computes a proper Beta–Binomial posterior only when an integer event count exists — never fabricated from a variance. Aggregate summaries then stream through a server-side proxy that holds the credential; a boundary check rejects patient-level identifiers first.

Value Model review runs only on cohorts that can support it; PHI and credential boundaries are enforced in code, not convention.

Integrity The PHI guard blocks record-level keys *and* scans string values for structured identifiers (email, SSN, phone, DOB), reporting the pattern and path but never the value. The audit trail reports only what was computed — the Beta prior is labelled *uncalibrated*, never claiming a calibration that did not happen.

Module	Decision it drives	Primary safeguard
<code>getClinicalFlags()</code>	Who needs outreach & why	Four-state absence inference; immutable thresholds
<code>stratifyPatients()</code>	Graded risk tier per patient	Computes on raw values; "unknown" when unmeasured
<code>assessCohortEvidence()</code>	Whether a cohort can be analysed	Precision + drift gating; honest, uncalibrated prior
<code>getLLMFeedback()</code>	Cohort-level model review	Key + value PHI scan; server-side credential boundary

Regression-guarded. A 22-check harness (`test_pipeline.mjs`) locks the four-state flag model, the tier ladder, the PHI key+value scan, gating monotonicity (more data never blocks a cohort), the no-unearned-provenance audit invariant, and the Beta credible interval against reference values — so a future edit fails loudly rather than drifting silently.